Experiment 2: Conditional Statements

Objective: To understand evaluation of prefix and postfix operators and implement decision-making constructs like if-else.

Questions:

1. **Write C programs to illustrate the use of unary prefix and postfix increment and decrement operators. Also, justify the answers.**

a. `int a = 5;`
`printf("%d %d %d", a++, ++a, a++);`
	b. `int x = 10;`
	`x = x++ + ++x;`
	`printf("%d", x);`

c. `int i = 3;`
`int j = i++ + ++i + i--;`
`printf("%d %d", i, j);`
	d. `int a = 4, b = 6;`
	`int c;`
	`c = a++ + --b + ++a;`
	`printf("a=%d b=%d c=%d", a, b, c);`

e. `int a = 0;`
`int b = 5;`
`if (a++ && b++)`
`   printf("%d %d", a, b);`
`else`
`   printf("%d %d", a, b);`
	f. `int a = 5, b = 5;`
	`if(a++ == ++b)`
	`   printf("Equal");`
	`else`
	`   printf("Not Equal");`

g. `int x = 5;`
`int y = ++x * x++ + --x;`
`printf("%d %d", x, y);`
	h. `int x = 1;`
	`if (++x || x++)`
	`   printf("%d", x);`

i. `int a = 2;`
`int b = (a++, ++a, a++);`
`printf("%d %d", a, b);`
	j. `int i = 2;`
	`if (i++ && i++)`
	`   printf("%d", i);`
	`else`
	`   printf("%d", i);`

k. `int a = 0, b = 3;`
`if ((a = 1) || b++)`
`   printf("%d %d", a, b);`
	l. `int x = 0, y = 1, z = 2;`
	`if (x++ || y++ && z++)`
	`   printf("%d %d %d", x, y, z);`

m. `int a = 0;`
`if (a && a++)`
`   printf("%d", a);`
`else`
`   printf("%d", a);`
	n. `int x = 0, y = 1;`
	`if (x && ++y)`
	`   printf("%d", y);`
	`if (x & ++y)`
	`   printf("%d", y);`
	`printf("%d", y);`

o. `int i = 1;`
`if (i++ || i++ && ++i)`
`   printf("%d", i);`
	p. `int x = 1;`
	`int y = -1;`
	`if (x == 50 || ++y) {`

	printf("if block executed\n");
	printf("Value of y: %d", y);
	}
	else {
	printf("else block executed\n");
	printf("Value of y: %d", y);
	}

2. **Write a C program to find the largest of three numbers using nested if-else statements.**
   Sample Input:
   `Enter three numbers: 10 55 23`
   Sample Output:
   `55 is the largest number.`

3. **Write a program to check if a given year is a leap year.**
   Sample Input:
   `Enter a year: 2024`
   Sample Output:
   `2024 is a leap year.`

4. **Write a program to check whether a character is a vowel or a consonant.**
   Sample Input:
   `Enter a character: E`
   Sample Output:
   `E is a vowel.`

5. **Create a simple calculator that performs addition, subtraction, multiplication, and division based on user input.**
   Sample Input:
   `Enter operator (+, -, *, /): *`
   `Enter two operands: 5 10`
   Sample Output:
   `5 * 10 = 50`

6. **Write a program to calculate an electricity bill based on the following slabs:**
   For the first 50 units: Rs. 0.50/unit
   For the next 100 units: Rs. 0.75/unit
   For the next 100 units: Rs. 1.20/unit
   For units above 250: Rs. 1.50/unit
   An additional surcharge of 20% is added to the bill.
   Sample Input:
   `Enter total units consumed: 200`
   Sample Output:
   `Total Electricity Bill: Rs. 165.00`